

packages/graph-core/src/__tests__/embedding-search.test.ts

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Overview

A new test file `packages/graph-core/src/__tests__/embedding-search.test.ts` (lines 1-92) has been added. It validates the `cosineSimilarity` helper and the `SemanticSearchEngine` class.

Key changes

- **Imports** (R1-R3): `vitest` helpers, `SemanticSearchEngine`, `cosineSimilarity`, and the `GraphNode` type.
- **Test data** (R5-R7): a mock `nodes` array of three `GraphNode` objects and a 4-dimensional `embeddings` map.
- **Unit tests** for `cosineSimilarity` covering identical, orthogonal, similar, and zero-vector cases.
- **Comprehensive tests** for `SemanticSearchEngine`:
 - Result ordering by similarity.
 - `limit` and `threshold` options.
 - `types` filtering.
 - Handling of missing embeddings.
 - `hasEmbeddings` and `addEmbedding` behavior.

Impact

- Adds coverage for embedding utilities; no production code is modified, so runtime behavior is unchanged.
- Provides concrete test scenarios that can serve as a reference for future changes to the search engine or similarity function.

Risks & follow-ups

- If `SemanticSearchEngine` does not implement the `types` filter or `threshold` logic as expected, the tests will fail; verify the implementation matches the test expectations.
- Ensure the relative import `../embedding-search.js` resolves correctly in the test environment.
- The test embeddings are 4-dimensional; confirm this matches the dimensionality used in production (unknown from the available diff/scan evidence).
- Run the full test suite to confirm no flakiness is introduced.